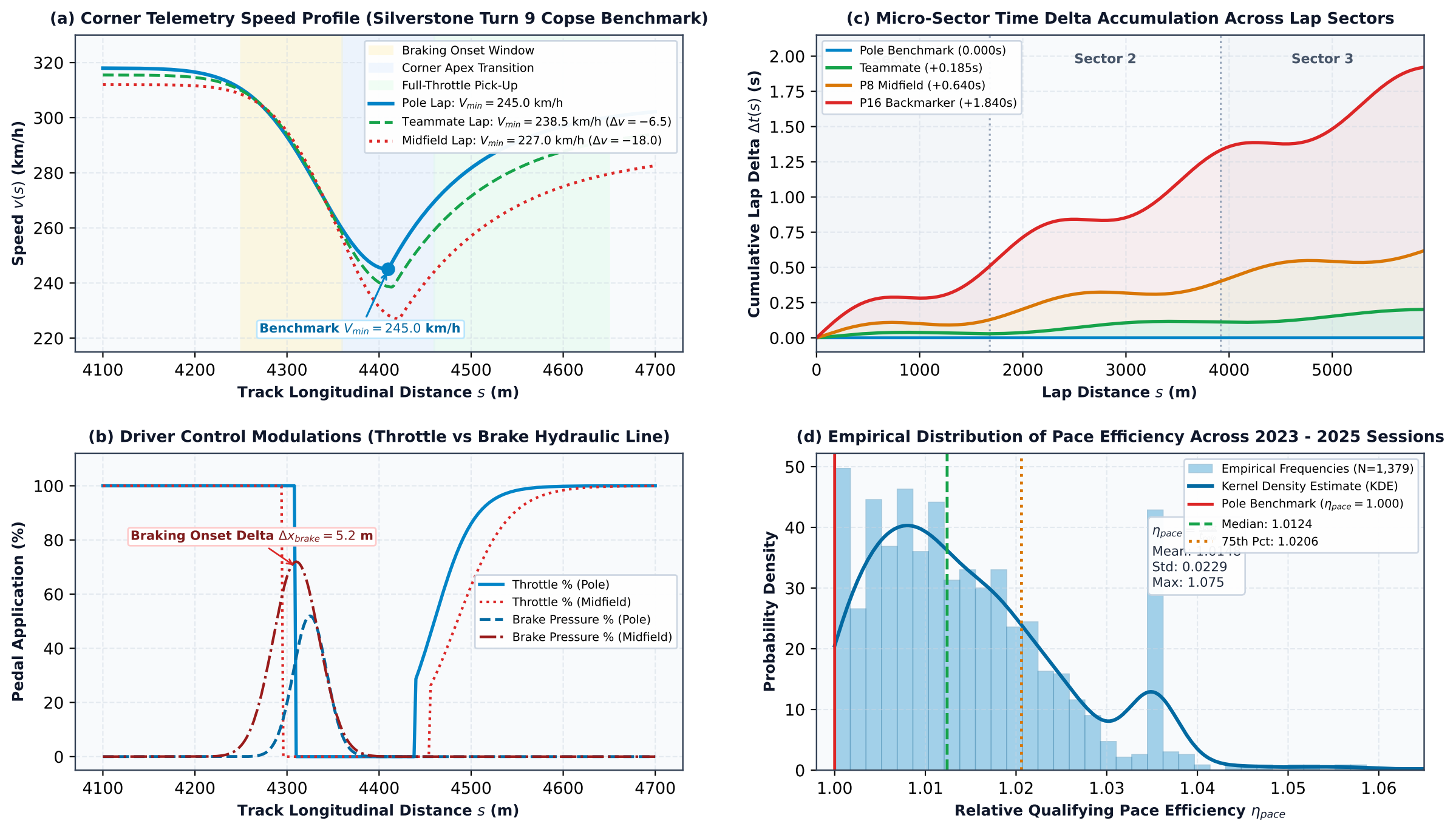

(a) Corner Telemetry Speed Profile (Silverstone Turn 9 Copse Benchmark)
Braking Onset Window
Corner Apex Transition
Full-Throttle Pick-Up
Pole Lap: $V_{min} = 245.0$ km/h
Teammate Lap: $V_{min} = 238.5$ km/h ($\Delta v = -6.5$)
Midfield Lap: $V_{min} = 227.0$ km/h ($\Delta v = -18.0$)
Benchmark $V_{min} = 245.0$ km/h
Speed $v(s)$ (km/h)
Track Longitudinal Distance s (m)
(b) Driver Control Modulations (Throttle vs Brake Hydraulic Line)
Braking Onset Delta $\Delta x_{brake} = 5.2$ m
Throttle % (Pole)
Throttle % (Midfield)
Brake Pressure % (Pole)
Brake Pressure % (Midfield)
Pedal Application (%)
Track Longitudinal Distance s (m)
(c) Micro-Sector Time Delta Accumulation Across Lap Sectors
Pole Benchmark (0.000s)
Teammate (+0.185s)
P8 Midfield (+0.640s)
P16 Backmarker (+1.840s)
Sector 2
Sector 3
Cumulative Lap Delta $\Delta t(s)$ (s)
Lap Distance s (m)
(d) Empirical Distribution of Pace Efficiency Across 2023 - 2025 Sessions
Empirical Frequencies (N=1,379)
Kernel Density Estimate (KDE)
Pole Benchmark ($\eta_{pace} = 1.000$)
Median: 1.0124
75th Pct: 1.0206
η_{pace}
Std: 0.0229
Max: 1.075
Probability Density
Relative Qualifying Pace Efficiency η_{pace}